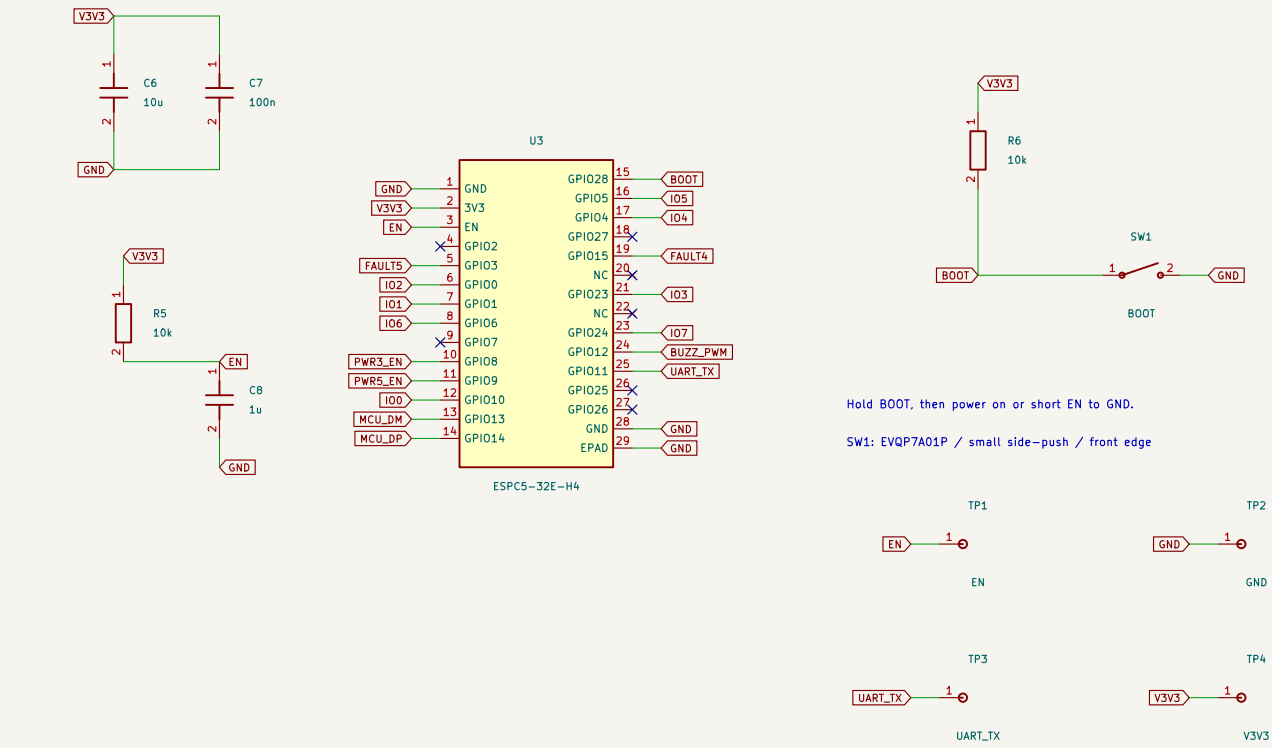
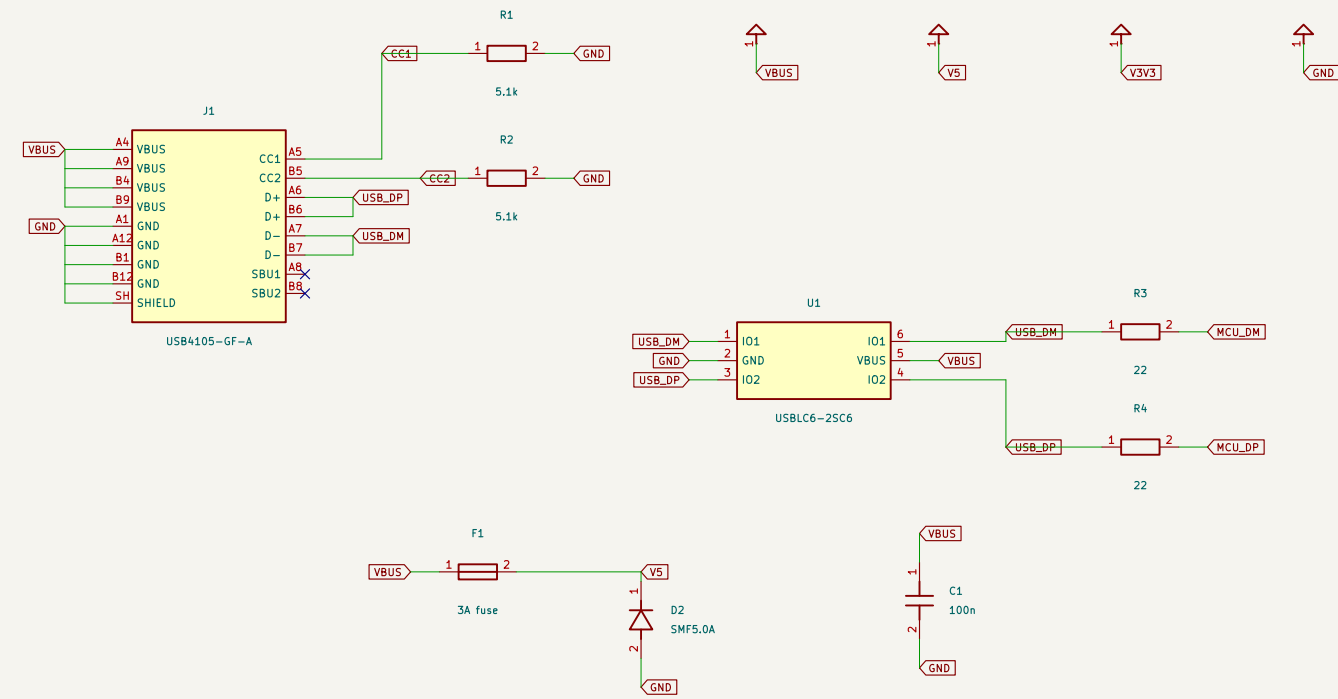


01 ESP32-C5 / BOOT / RESET

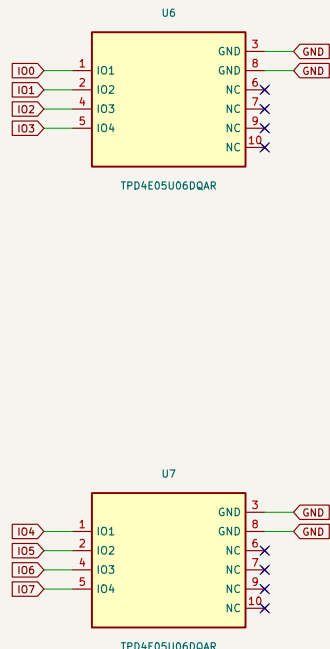


External dual-band U.FL antenna; 3.3 V GPIO only.

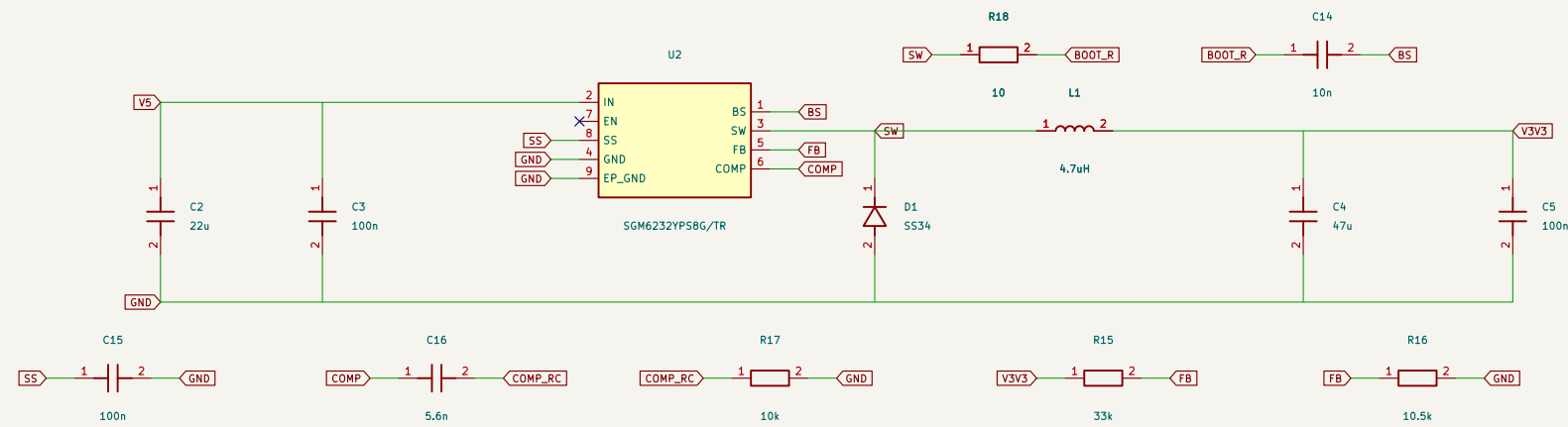
02 USB-C / INPUT PROTECTION



07 HEADER ESD / SHUNT PROTECTION

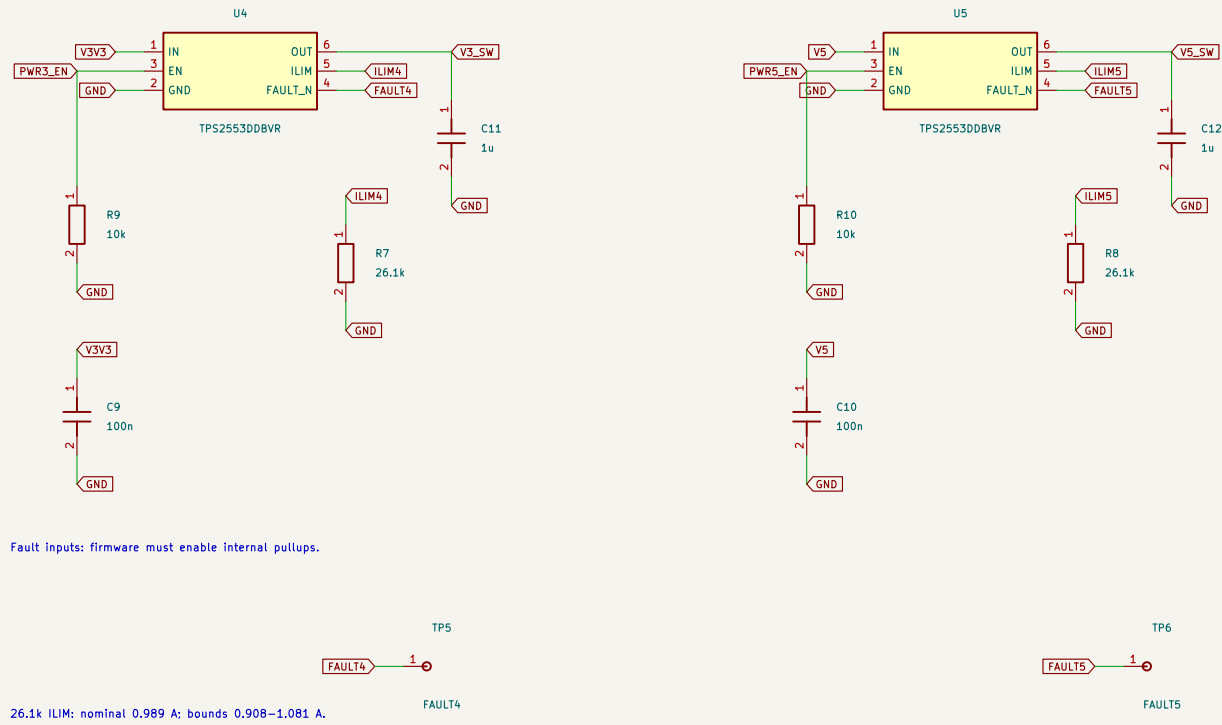


03 SGM6232 / ALWAYS-ON 3.3 V



EN open: auto-start. EP to GND. FB divider 33k / 10.5k = 3.314 V.

04 SWITCHED OUTPUTS / 1 A NOMINAL LIMIT



FAULT4 -> Input 15; FAULT5 -> Input 3. Active LOW.

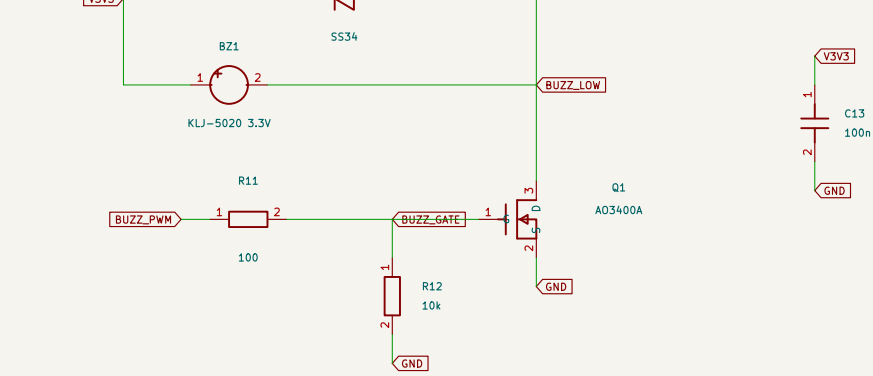
PROTOTYPE VALIDATION

Check startup, dropout, ripple, heat.

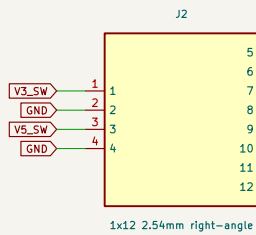
Use specified 5 V / 3 A adapter.

Verify current limit and both loaded rails.

05 PASSIVE 5020 BUZZER



06 12-PIN MALE HEADER



U6/U7: four channels each; no supply or bypass needed.

I00-I07: 3.3 V only; final-board ESD test required.

U10: switched rails. Power current bypasses the array.

Place arrays and bypass capacitors directly beside J2.

Short, wide ground returns: qualify ESD on final PCB.

48 x 28 mm PCB - top-side component assembly.